Abhayrajsinh Parmar

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Technical Skills

Languages:	Python, JavaScript, SQL
AI & LLM:	Retrieval-Augmented Generation (RAG), Generative AI, Prompt Engineering, LLM Agents, Model Context Protocol (MCP), Structured Output Validation (Pydantic), Rule-Based Expert Systems (experta), LLM Evaluation
Machine Learning:	scikit-learn, Classification / Regression, Cross-Validation, Precision / Recall / F1
Retrieval & Vector Search:	Qdrant, Hybrid Search (Dense + Sparse BM25 + ColBERT Late Interaction), Embeddings, LangChain
LLMs & Serving:	vLLM (continuous batching, PagedAttention), OpenAI API, Groq, Hugging Face, Ollama, Quantization, MoE
Backend & APIs:	FastAPI (Async / ASGI), REST APIs, SSE Token Streaming, asyncio Concurrency Control, Node.js / Express
Data & Documents:	PyMuPDF, pdfplumber, Tesseract OCR, BeautifulSoup, tiktoken, Pandas, NumPy
Databases:	MongoDB (Motor), PostgreSQL, MySQL
DevOps & Cloud:	Docker, AWS, RunPod, Git / GitHub

Experience

AI/ML Engineer | Maven Profcon Services LLP, Ahmedabad, India — Aug 2025 – Sep 2026

- One of 3 engineers building a shared AI platform serving 8 production LLM / Generative-AI modules — regulatory chatbot, document Q&A, resume ranking, WhatsApp agent, medical-device classification and FDA / CDSCO ingestion — behind 28 REST endpoints on an asynchronous FastAPI backend, used by 60–100 internal staff plus client-facing deployments.
- Architected the shared retrieval infrastructure reused across every module: a three-stage hybrid Qdrant pipeline — dense semantic (qwen3-embedding, 4096-dim) and sparse BM25 each prefetching 300 candidates, reranked by jina-colbert-v2 late interaction — with payload-filtered multi-tenancy isolating 11 knowledge domains across 3 collections.
- Built a hybrid neuro-symbolic classifier for EU MDR / IVDR medical-device risk class: selected a 76-rule forward-chaining expert system (experta) for the regulated determination — deterministic and auditable rather than model-generated — behind an LLM-driven question flow; 100+ devices classified to date, spot-checked at 92–96% agreement with in-house regulatory experts.
- Engineered a resumable FDA 510(k) ingestion pipeline (asyncio, vLLM, PyMuPDF, Tesseract) that processed 175,000+ documents in ~6 weeks at ~30–50s per document under 8-way concurrency; ~35% required OCR fallback and ~25% hit a retryable failure, absorbed by a per-document status machine with per-error retry budgets and idempotent upserts so interrupted runs resumed.
- Led the vLLM migration to self-hosted open-weight and Mixture-of-Experts (MoE) models on rented RunPod GPUs — weighing VRAM footprint, quantization, active-vs-total parameters, context length and throughput — replacing per-token API spend with fixed GPU cost.
- Enforced production reliability: pure-ASGI SSE-safe auth middleware, Pydantic validation, system-prompt guardrails, exponential-backoff retries, structured logging, a /health endpoint fanning out to 6 dependencies, and CPU-bound PDF work isolated off the event loop.

Backend Developer | ADS Vizion, Ahmedabad, India — Jun 2024 – Jun 2025

- Built accounting and ERP backend in Node.js, Express and MySQL: invoicing, e-invoice / e-GST, financial reporting and transactions.
- Built a shipping-management module — load / unload jobs, party and supplier masters, cargo and vehicle records, role-based access.

Key Projects

PDF Summarization & Chatbot | *FastAPI · Qdrant · OpenAI · RAG · OCR*

- Document Q&A with hybrid RAG — dense Qdrant search fused with sparse BM25 via asyncio.gather, deduplicated and capped at 12 chunks per prompt; token-by-token streaming over FastAPI StreamingResponse with per-document history in MongoDB.
- Engineered OCR ingestion (PyMuPDF + Tesseract) with LangChain recursive chunking at 35K-character targets and 700-character overlap, per-chunk embedding batched 32 at a time under an asyncio semaphore, and TTL-cached BM25 indexes to avoid rebuilds.
- Exposed the full document lifecycle across 8 REST endpoints — extraction, summaries, cross-document search, rename and delete — used by 40–50 internal staff over several hundred documents.

Regulatory Chatbot with MCP | *FastMCP · MCP · Qdrant · OpenAI*

- Gave the LLM tool access through a Model Context Protocol server (FastMCP, streamable-HTTP) consumed via an MCP ClientSession.
- Implemented the retrieval stack — chunking, embedding, Qdrant search, medical-device classification, content extraction, and a query combiner that rewrites follow-ups into standalone queries.
- Indexed the complete published EU MDR and IVDR corpus — regulation text, annexes, classification rules and device code tables — across 6 payload-filtered domains.

Resume Ranking Engine | *FastAPI · OpenAI · MongoDB · Pydantic*

- Scored 2,500+ resumes against 10–12 job descriptions in a 2–3 week backfill of the candidate database, under 3 weight profiles by level (intern / fresher / experienced); now in production use by HR.
- Combined batched LLM calls — 10 candidates per request under a 5-way concurrency limit — for skill matching and experience extraction, with deterministic scoring for degree level, recency and gaps.
- Ingested resumes from URLs with file-type detection, PDF / DOCX parsing and Tesseract OCR fallback.

WhatsApp AI Agent | *FastAPI · Qdrant · OpenAI · OCR*

- Built a multimodal assistant ingesting text, PDFs and images (Tesseract OCR), merged into one structured context bundle per turn.
- Added LLM query rewriting for follow-ups, retrieval-grounded responses, rolling summarization and per-user history in MongoDB.
- Deployed on the company's WhatsApp Business line as first touch for domain questions, vacancy enquiries and new-business leads.

Education

B.E. in Computer Science & Engineering | New LJ Institute of Engineering and Technology | **CGPA: 9.36 / 10** — Oct 2020 – Jun 2024